
SPACE
VIGIL
Crew health, self-assessed. Decision support for astronauts on long missions.
Les Tigres de Bethesda
NASA Space Apps Challenge 2026 · Create Health Monitoring Software for Astronauts on Space Missions

24 MINUTES TO MARS

On a Mars transit, a question to the flight surgeon takes up to 48 minutes round trip. The crew has to assess and act first; the ground confirms later.

4-24 MIN

ONE-WAY DELAY

Signal time to Mars, NASA Ames

5

HAZARDS

Radiation, isolation, distance, gravity, closed cabin (NASA HRP)

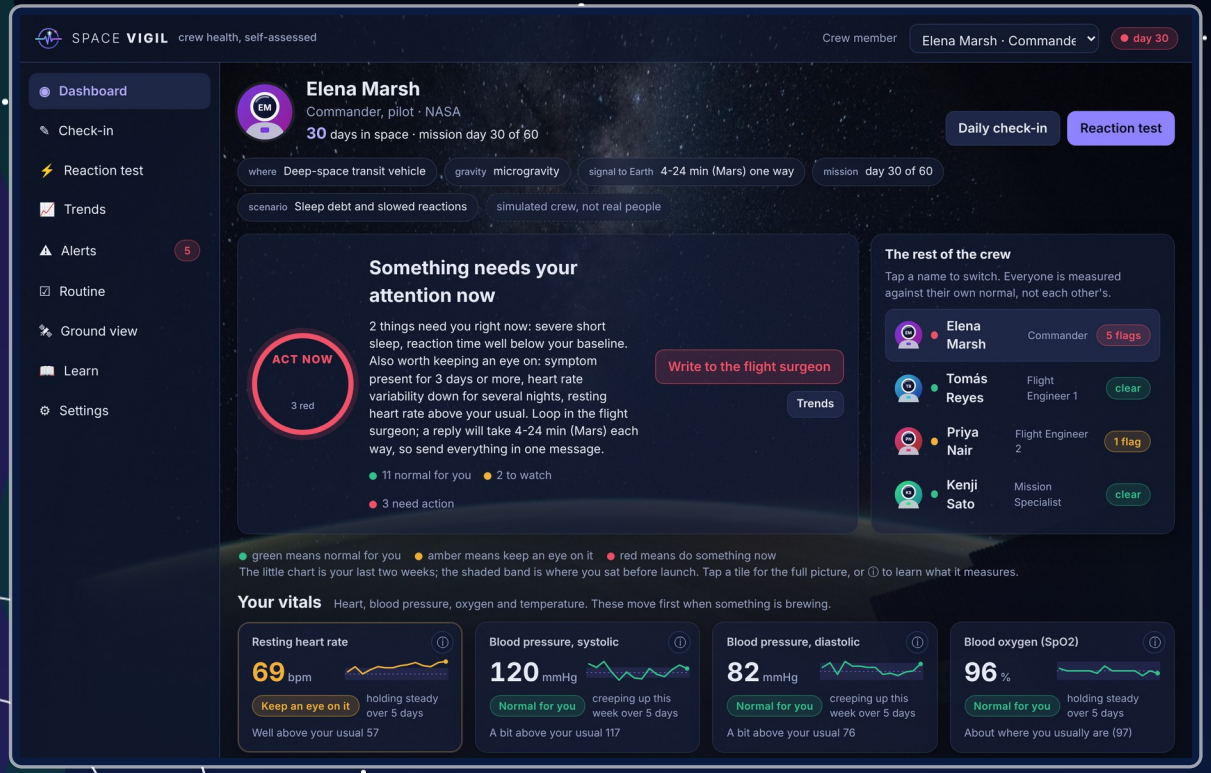
6.1 H

SLEEP ON ISS

Against 8.5 h scheduled (Barger 2014)

Each hazard shows up first as a small shift in an everyday number: a shorter night, a slower reaction, a few extra beats per minute at rest. Crews need software that notices those shifts and says what to do.

ONE SCREEN, ONE ANSWER



WHO

Portrait, name, role, agency, days in space. Everyone is judged against their own baseline.

16 SIGNALS

Heart, HRV, breathing, oxygen, temperature, pressure, glucose, mass, sleep, exercise, mood, stress, reaction time, lapses, dose.

THREE COLOURS

Normal for you, keep an eye on it, needs action. One ring sums it up.

JUDGED AGAINST YOU, NOT THE AVERAGE

YOUR BASELINE

14 pre-flight days give a median and a typical spread per signal. The shaded band on the chart is that baseline.

PUBLISHED LIMITS

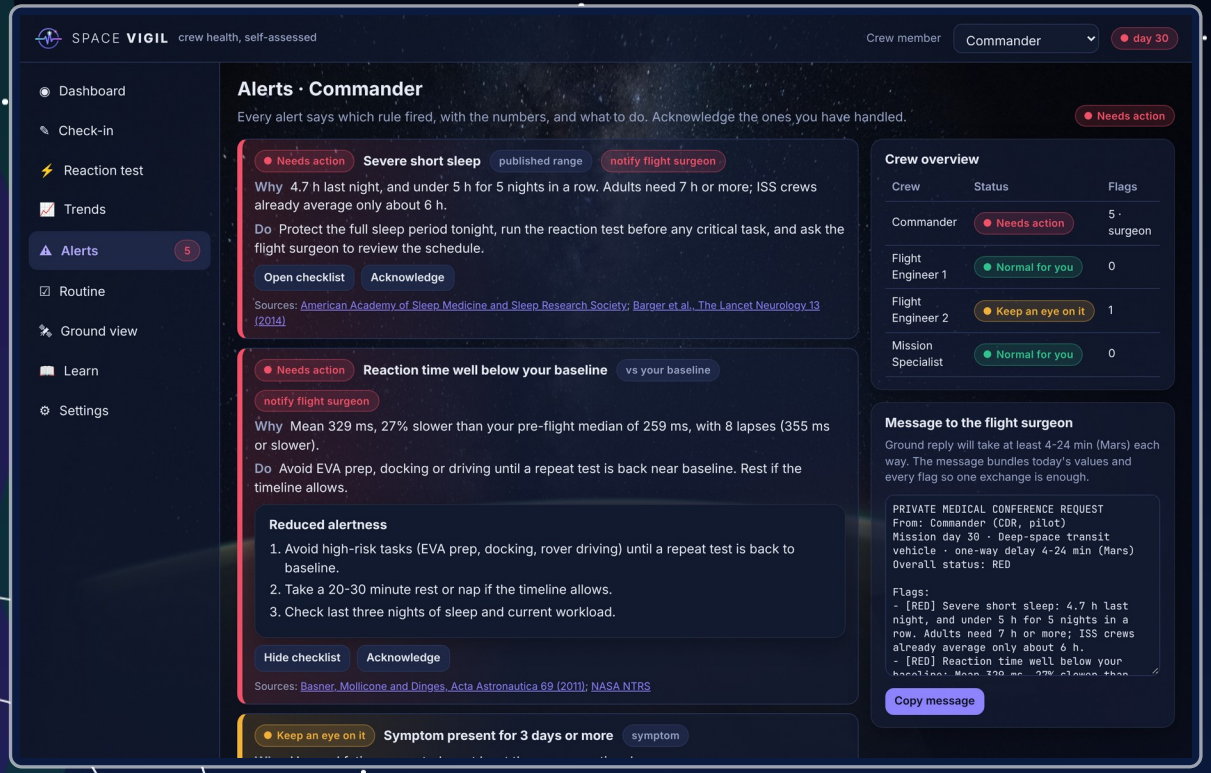
ACC/AHA, CDC, ADA, AASM and NASA-STD-3001 cut-offs drawn on the same chart.

WHY BOTH

A resting heart rate of 95 is normal for most adults and a real change for someone who sits at 55.



EVERY ALERT EXPLAINS ITSELF



WHY

The rule, the numbers and the source behind every flag.

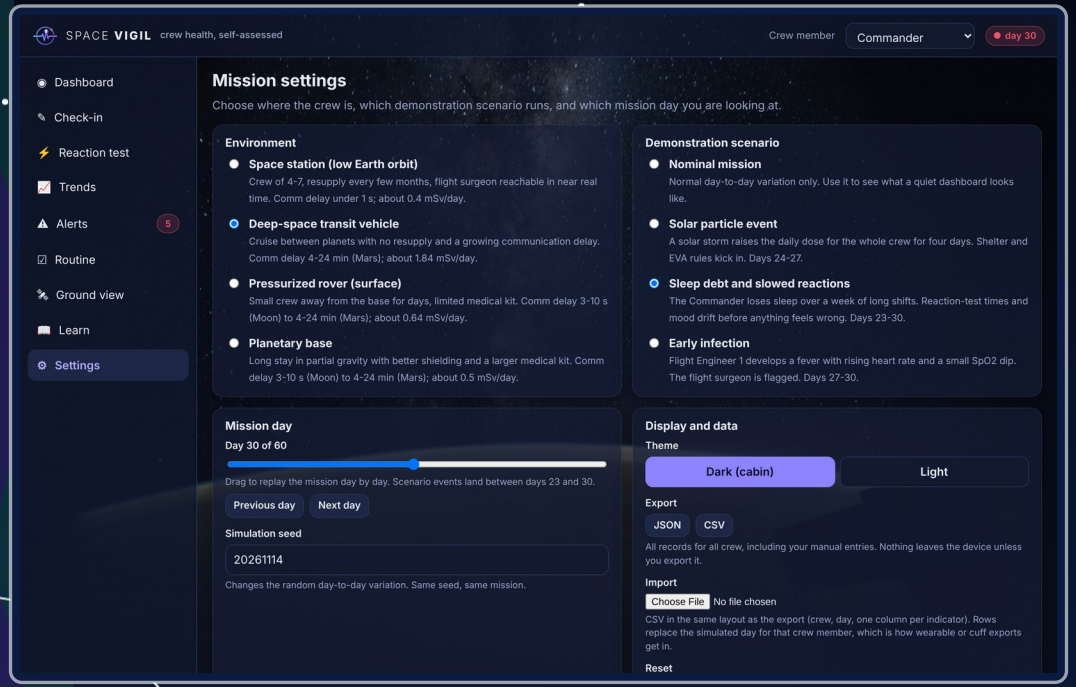
DO

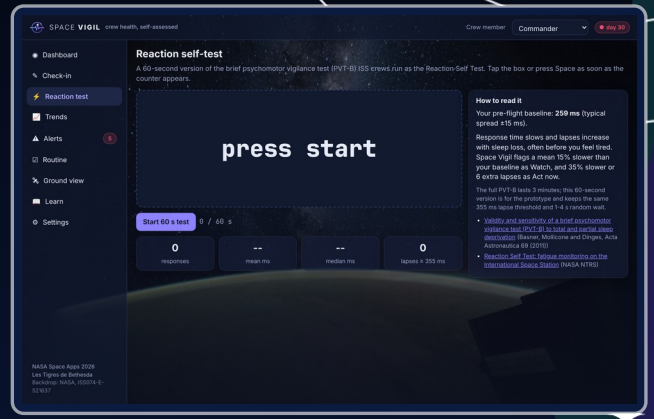
A checklist of next steps. Alerts that need the ground say so.

ONE MESSAGE

The whole day packaged for the flight surgeon, so one exchange is enough across a 48-minute round trip.

BUILT FOR THE WHOLE MISSION





Station, transit, rover or base, with measured comm delays and dose rates.

Four scenarios; a slider replays any mission day.

60-second reaction test and camera pulse.

Ground view for the flight surgeon.

Offline, installable, exportable.

WHAT IS REAL, AND WHAT IS NEXT

REAL

- Every threshold cites NASA-STD-3001, NASA HRP, NASA NTRS, MSL/RAD, ACC/AHA, CDC, ADA, AASM or a peer-reviewed paper.
- Comm delays and daily doses are measured values.
- Crew data is simulated and labelled as such; no real astronaut records.
- Decision support, not diagnosis.

NEXT

- Live Bluetooth feed from the Bio-Monitor (Astroskin) shirt already flying on the ISS.
- Baselines informed by NASA OSDR human datasets.
- Sync between crew tablets and the ground.
- Validate the 60-second reaction test against the 3-minute PVT-B.

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